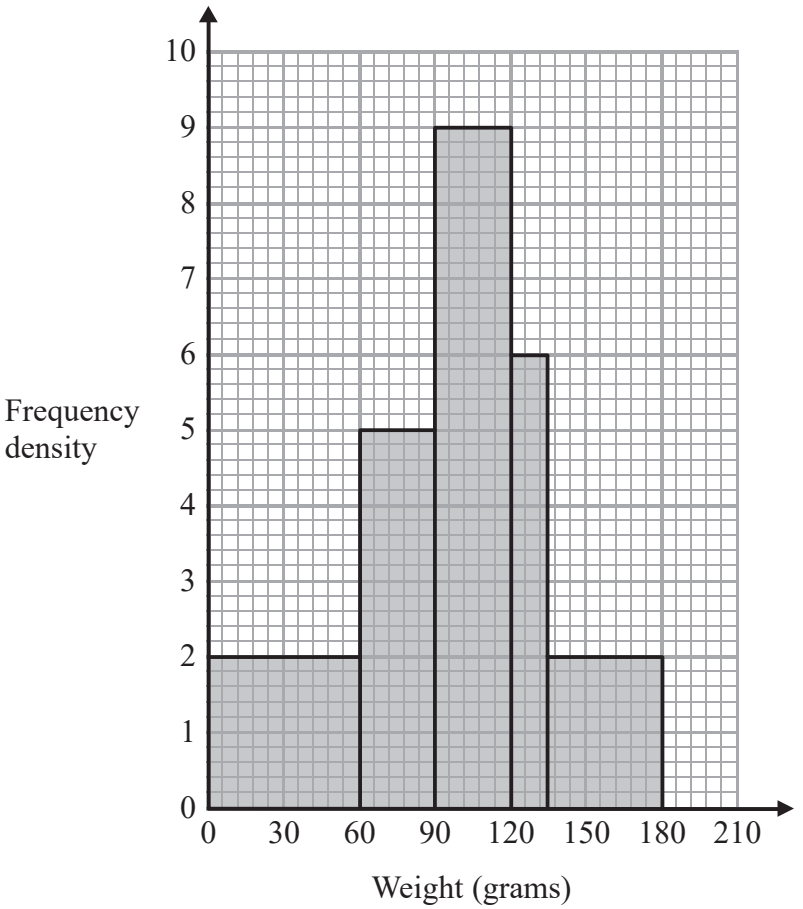
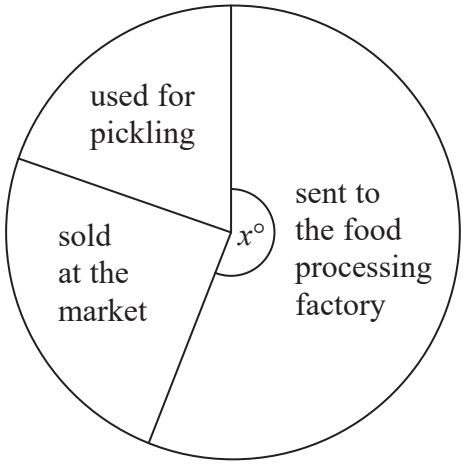
21 The histogram gives information about the distribution of the weights of some onions grown by a farmer.



Onions less than 60 grams in weight are used for pickling.
Onions greater than 120 grams in weight are sold at the market.
The rest of the onions are sent to a food processing factory.

A pie chart is drawn using the information opposite to show what the farmer does with the onions he grows.



The angle of the sector for the onions sent to the food processing factory is $x°$.

Work out the value of x.

Frequency = width × density.

1 mark
Read widths × densities.

0–60: 120; 60–90: 150; 90–120: 270.

120–135: 90; 135–180: 90.

1 mark
Total 720 onions.

Total = 720; factory (60–120) = 420.

1 mark
Factory frequency 420.

Sector = (420/720) × 360° = 210°.

1 mark
Angle x = 210°.

$x =$ 210

(Total for Question 21 is 4 marks)